# Modeling and Forecasting Monthly Atmospheric $CO_2$ at Mauna Loa, 1958–2026

A Seasonal ARIMA Analysis

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STA6938 Time Series · Final Project

# Roadmap

1. Background, motivation and data
2. Trend, seasonality and stationarity
3. Model identification and selection
4. Estimation, tests and confidence intervals
5. Model adequacy and remedial measures
6. Forecast evaluation and forecasts to 2030

**Research questions**

- How fast is $CO_2$ rising?
- Is the rise accelerating?
- What levels should we expect by 2030?

# Background and motivation

## Rationale

- $CO_2$ is the principal driver of anthropogenic climate change.
- Climate targets depend on *how fast* $CO_2$ rises and *when* it crosses thresholds.
- Rigorous characterization and forecasting are of direct scientific and policy relevance.
- Methodologically rich: stochastic trend, seasonal cycle and serial dependence.

## The question

$CO_2$ is rising; the question is **how fast**, **whether it is accelerating**, and **what to expect next**, with proper uncertainty.

## Objectives

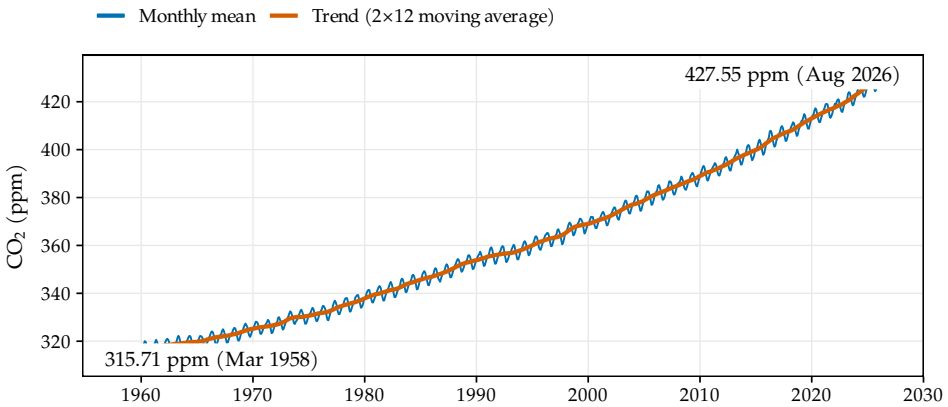
1. Characterize the trend and seasonality.
2. Build and validate an adequate model.
3. Test whether growth is accelerating.
4. Forecast to 2030 with prediction intervals.

# The data: Mauna Loa monthly $CO_2$

- Mauna Loa Observatory, Hawaii: the longest continuous direct $CO_2$ record, started by C. D. Keeling in 1958.
- **Source:** NOAA Global Monitoring Laboratory [2], https://gml.noaa.gov/ccgg/trends/data.html
- **Variable:** monthly mean mole fraction, in parts per million (ppm).
- Real, public and never analyzed by me before; a large sample for time series modeling.

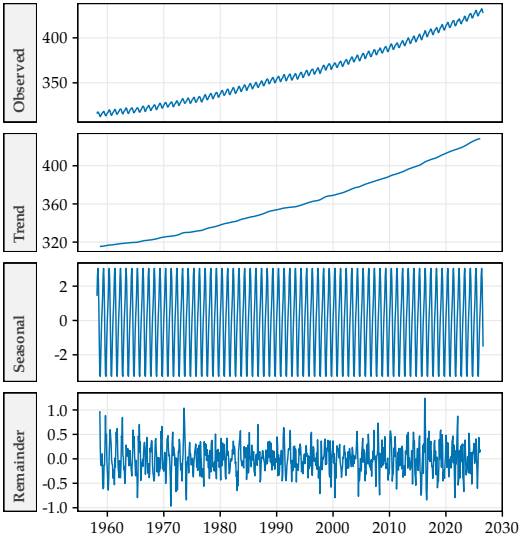
| The series at a glance | |
|---|---|
| Months, $n$ | 822 |
| Period | 1958–2026 |
| Missing values | none |
| First value | 315.71 ppm |
| Last value | 427.55 ppm |
| Increase | +111.8 ppm |
| Relative increase | 35% |

# Step 1: the time plot




**Takeaway:** An upward trend and a regular 12-month cycle: the mean depends on $t$, so the series is non-stationary.

# Decomposition: trend plus a stable seasonal cycle

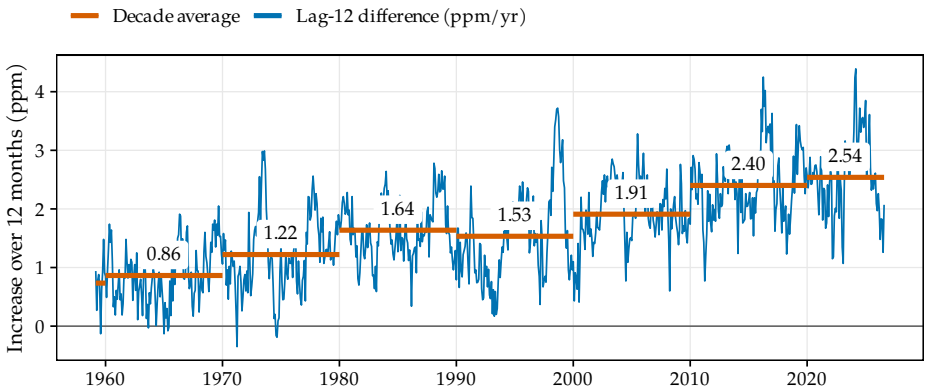



- Additive, $2 \times 12$ moving-average filter.
- Trend: 96.6% of the variance.
- Peak in **May** ($+3.04$ ppm), trough in **October** ($-3.25$ ppm); amplitude **6.29 ppm**.
- Amplitude 6.0 ppm (1960s) vs 7.0 ppm (2010s), while the level rose 35%.

**Decision**

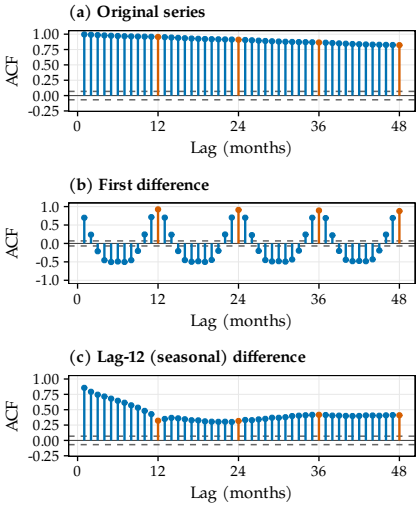
Swings do not grow with the level $\Rightarrow$ **additive model, no transformation**.

# The growth rate is accelerating




**Takeaway:** Growth tripled: 0.86 ppm/yr (1960s) → 2.54 ppm/yr (2020s). Record jumps in El Niño years (2016, 2024).

# Achieving stationarity



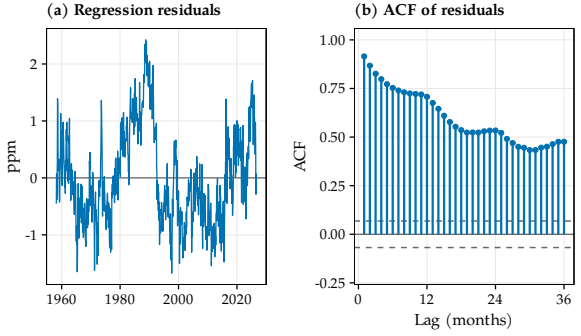

- $X_t$: ACF decays very slowly ($\hat{\rho}(1) = 0.996$).
- $\nabla X_t$: ACF $\approx 0.9$ at lags 12, 24, 36 $\Rightarrow$ seasonal difference.
- $\nabla_{12} X_t$: slow decay $\Rightarrow$ regular difference.

**Augmented Dickey–Fuller**, $H_0$: unit root

| Series | Type | $\tau$ | Decision |
|---|---|---|---|
| Seas. adj. $X_t$ | mean | 3.57 to 4.85 | keep $H_0$ |
| Seas. adj. $X_t$ | trend | $-0.88$ to $-0.62$ | keep $H_0$ |
| $\nabla\nabla_{12} X_t$ | mean | $-37.9$ to $-22.1$ | reject |

$\Rightarrow$ $W_t = (1 - B)(1 - B^{12})X_t$ is **stationary**.

# Estimation versus differencing



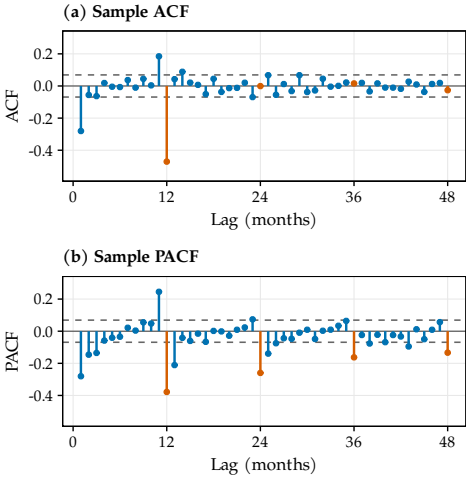

Quadratic trend + 11 monthly dummies:

- Fit: $R^2 = 0.9994$.
- Residual ACF(1) $= 0.915$; $Q(12) = 5991$.
- Durbin–Watson $D = 0.17$ ($\ll 2$).
- ARMA errors: AR root 1.02.

**Conclusion**

Deterministic trend fails: **difference instead**.

# Identification from the ACF and PACF of $W_t$




- $Q(12) = 283.3$, $p < 0.001$: $W_t$ is not white noise.
- **ACF cuts off** after lags 1 ($-0.28$) and 12 ($-0.47$).
- **PACF tails off** at lags 1–3 and 12, 24, 36.
- $\Rightarrow$ MA(1) $\times$ seasonal MA(1).

**ARIMA**$(0,1,1) \times (0,1,1)_{12}$

$$(1-B)(1-B^{12})X_t = (1-\theta_1 B)(1-\Theta_1 B^{12})Z_t$$

# Model selection: AIC and SBC

| Model | Parameters | AIC | SBC | ΔSBC | Ljung–Box $Q(24)$ $p$ |
|---|---|---|---|---|---|
| **ARIMA**$(0,1,1)\times(0,1,1)_{12}$ | **2** | 451.30 | **465.39** | **0.00** | 0.532 |
| ARIMA$(1,1,1)\times(0,1,1)_{12}$ | 3 | **447.58** | 466.37 | 0.98 | 0.769 |
| ARIMA$(0,1,2)\times(0,1,1)_{12}$ | 3 | 448.59 | 467.38 | 1.99 | 0.723 |
| ARIMA$(0,1,1)\times(1,1,1)_{12}$ | 3 | 453.30 | 472.08 | 6.69 | 0.472 |
| ARIMA$(2,1,0)\times(0,1,1)_{12}$ | 3 | 460.64 | 479.42 | 14.03 | 0.213 |
| ARIMA$(1,1,0)\times(0,1,1)_{12}$ | 2 | 472.83 | 486.92 | 21.53 | 0.038 |
| ARIMA$(0,1,1)\times(1,1,0)_{12}$ | 2 | 669.34 | 683.43 | 218.04 | $< 0.001$ |

- Nine candidates fitted by maximum likelihood; **SBC** preferred in this large sample.
- Pure AR and seasonal-AR alternatives fit clearly worse.

# Four model families compared

| Family | Best model (by SBC) | ACF lag 12 | $Q(12)$ $p$ | White noise? |
|---|---|---|---|---|
| ARMA | ARMA(3,2) for $W_t$ | $-0.409$ | $< 0.001$ | No |
| ARIMA | ARIMA(4,1,4) with drift | 0.180 | $< 0.001$ | No |
| Regression + ARMA errors | ARMA(2,1) errors | 0.138 | $< 0.001$ | No |
| **Seasonal ARIMA** | **ARIMA**$(0,1,1) \times (0,1,1)_{12}$ | $\mathbf{-0.005}$ | **0.464** | **Yes** |

- Non-seasonal ARIMA mimics the seasonal difference with AR roots *on* the unit circle (periods 6 and 12).
- Only the seasonal ARIMA captures the seasonal structure, with just **two parameters**.

# Estimates, hypothesis tests and confidence intervals

| Method | Parameter | Estimate | SE | $z$ | $p$ | 95% CI |
|---|---|---|---|---|---|---|
| Maximum likelihood | $\theta_1$ | 0.3885 | 0.0367 | 10.59 | $< 0.001$ | [0.317, 0.460] |
| | $\Theta_1$ | 0.8700 | 0.0170 | 51.28 | $< 0.001$ | [0.837, 0.903] |
| Conditional least squares | $\theta_1$ | 0.4172 | 0.0360 | 11.60 | $< 0.001$ | [0.347, 0.488] |
| | $\Theta_1$ | 0.8116 | 0.0194 | 41.84 | $< 0.001$ | [0.774, 0.850] |

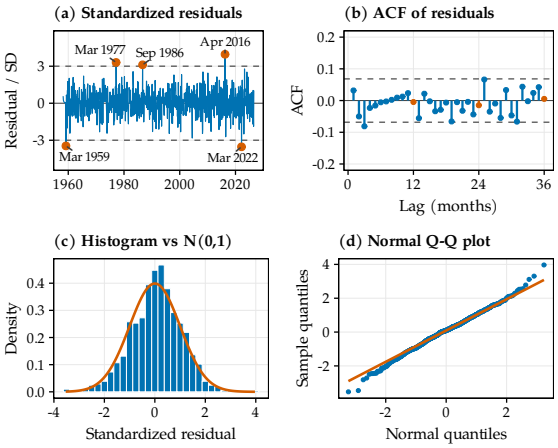
**Inference**

Both parameters highly significant; ML and CLS agree. $\hat{\sigma}_Z = 0.318$ ppm.
Invertible: roots 2.57 and $1.012 > 1$.

**Interpretation**

$\hat{\theta}_1$: 39% of a monthly shock reverses next month.
$\hat{\Theta}_1 = 0.87$: very stable seasonal pattern; CI excludes 1.

# Model adequacy: residual diagnostics




| Test | Statistic | $p$ |
|---|---|---|
| Ljung–Box $Q(6)$ | 9.05 | 0.060 |
| Ljung–Box $Q(12)$ | 9.74 | 0.464 |
| Ljung–Box $Q(18)$ | 14.43 | 0.567 |
| Ljung–Box $Q(24)$ | 20.83 | 0.532 |
| $Q(12)$, squared resid. | 12.40 | 0.414 |
| Shapiro–Wilk | 0.996 | 0.048 |

- Independence ✓ Constant variance ✓
- Normality marginal: 5 months beyond $\pm 3$.

# Remedial measures

## 1. Small lag-3 autocorrelation

Added an AR(1) term: $\hat{\phi}_1 = 0.215$ ($p = 0.010$).

- SBC rises (466.37 vs 465.39).
- Hold-out RMSE worse: 1.168 vs 1.110.
- Interval coverage only 83% vs 97%.

$\Rightarrow$ **Rejected**: keep the parsimonious model.

## 2. Five outlying months

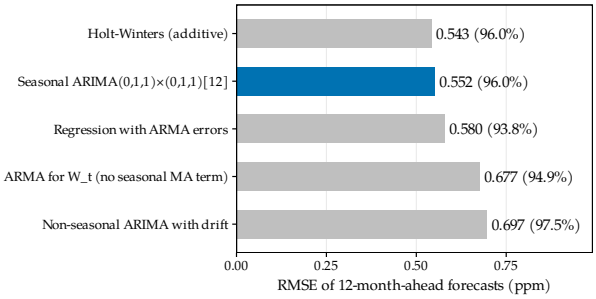
Re-estimated with those months set to missing:

- $\hat{\theta}_1 = 0.384$, $\hat{\Theta}_1 = 0.869$: **unchanged**.
- Normality restored: Shapiro–Wilk $p = 0.211$, Anderson–Darling $p = 0.101$.

$\Rightarrow$ Inference is **robust**.

**Takeaway:** ARIMA$(0,1,1) \times (0,1,1)_{12}$ is adequate; no transformation is needed.

# Forecast evaluation

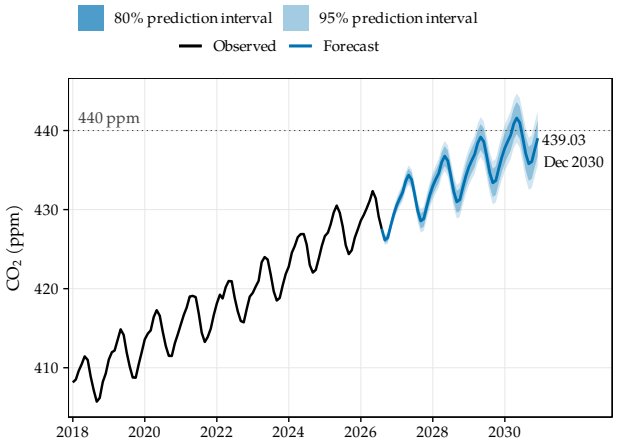



**Rolling origin:** refit each August 2000–2022; forecast 12 months (276 forecasts).

- Seasonal ARIMA: RMSE 0.552 ppm, 95% coverage 96.0%.
- Best of the four model families; tied with Holt–Winters smoothing.

**Hold-out** (Sep 2023 – Aug 2026): RMSE 1.110 ppm, coverage 97.2%.

# Forecasts to 2030




## April 2030

$CO_2$ first exceeds **440 ppm**

440.84 ppm [437.78, 443.89]

- Dec 2030: 439.03 ppm
  95% PI [435.59, 442.47]
- Growth $\approx$ **2.4 ppm/yr**
- Holt–Winters agrees within 0.13 ppm.

# Conclusions

1. $CO_2$ is non-stationary with an **accelerating trend** and a **stable 12-month cycle**; regular and seasonal differencing achieve stationarity.
2. **ARIMA**$(0,1,1) \times (0,1,1)_{12}$ is selected by ACF/PACF identification and SBC; both parameters are highly significant and the model is invertible.
3. Residuals are white noise with constant variance; inference is robust to five outlying months.
4. Best of four model families in out-of-sample forecasting (RMSE 0.55 ppm, 96% coverage).
5. $CO_2$ rises $\approx 2.4$ ppm/yr and is forecast to pass 440 ppm in April 2030.

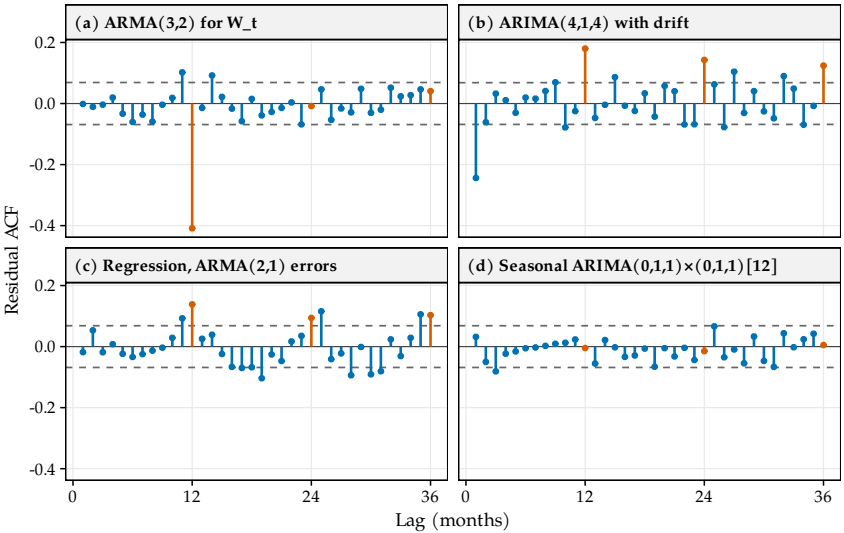
**Limitations and extensions**

Univariate model: it cannot anticipate El Niño surges (2016, 2024). Adding climate indices or emissions data is a natural extension.

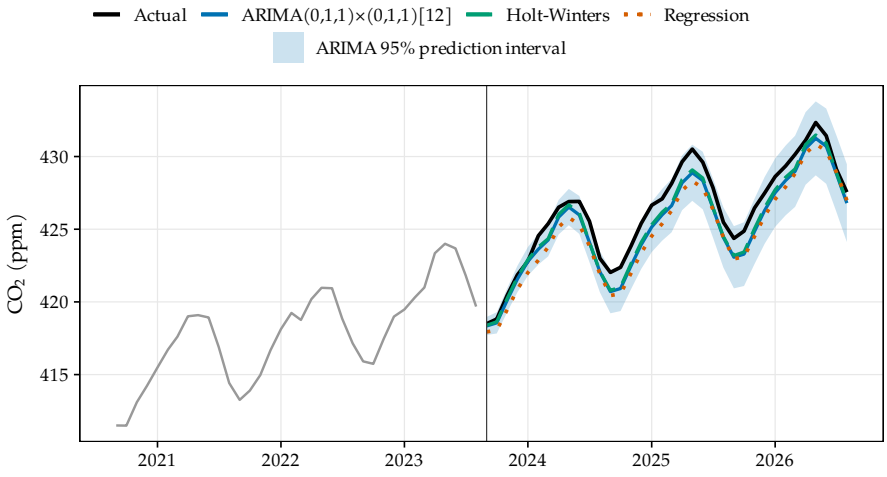
# Thank you

**Questions?**

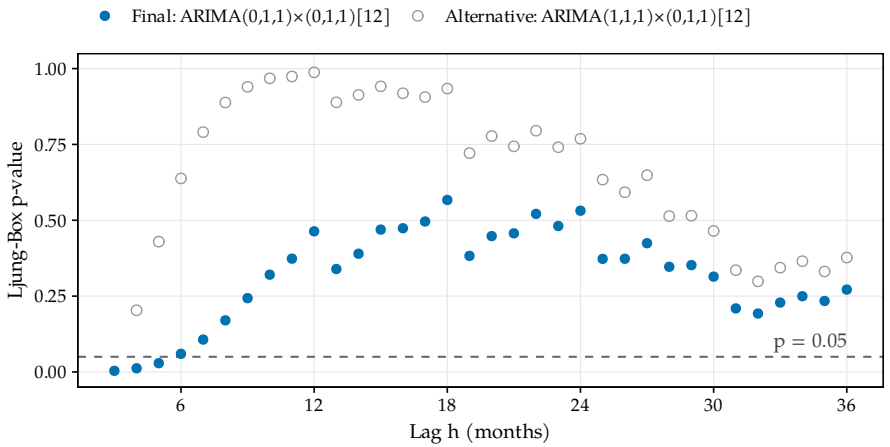
# Backup: residual ACF of the four model families



# Backup: 36-month hold-out forecasts



# Backup: Ljung–Box $p$-values by lag



# References


[1] C. D. Keeling (1960). The concentration and isotopic abundances of carbon dioxide in the atmosphere. *Tellus*, 12(2), 200–203. doi:10.1111/j.2153-3490.1960.tb01300.x

[2] X. Lan & R. F. Keeling (2026). Trends in atmospheric carbon dioxide: Mauna Loa monthly mean data. NOAA Global Monitoring Laboratory. https://gml.noaa.gov/ccgg/trends/data.html

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[4] D. A. Dickey & W. A. Fuller (1979). Distribution of the estimators for autoregressive time series with a unit root. *JASA*, 74(366), 427–431. doi:10.1080/01621459.1979.10482531

[5] G. E. P. Box, G. M. Jenkins, G. C. Reinsel & G. M. Ljung (2015). *Time Series Analysis: Forecasting and Control* (5th ed.). Wiley. ISBN 978-1-118-67502-1

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